Finite Element Method

FEM for Multidimensional Vector Field Problems – Part 1
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- This lecture covers finite element solutions for linear elasticity problems.

- The theory of linear elasticity hinges on the following four assumptions:
  1. deformations are small;
  2. the behaviour of the material is linear;
  3. dynamic effects are neglected;
  4. no gaps or overlaps occur during the deformation of the solid.

- The requirements of a linear stress analysis solution are closely related to the assumptions:
  1. the body must be in equilibrium;
  2. it must satisfy the stress–strain law;
  3. the deformation must be smooth.

- In addition to the above, in order to write a stress–strain law, we need a measure of the strain that expresses the strain in terms of the deformation, which is called the **strain–displacement equation**.

- The displacement vector in two dimensions is a vector with two components. Using a Cartesian coordinate system, the displacement components are the x- and y-components, written in matrix and vector forms as

$$\mathbf{u} = \begin{bmatrix} u_x \\ u_y \end{bmatrix}, \qquad \vec{u} = u_x\,\vec{i} + u_y\,\vec{j}, \tag{1}$$

  where the subscript indicates the component.

- In finite element methods, the strains are usually arranged in a column matrix $\boldsymbol{\varepsilon}$:

$$\boldsymbol{\varepsilon} = \begin{bmatrix} \varepsilon_{xx} & \varepsilon_{yy} & \gamma_{xy} \end{bmatrix}^{\mathrm{T}}. \tag{2}$$

- Equations (9.2)–(9.3) can be written in terms of the displacements as a single matrix equation:

$$\boldsymbol{\varepsilon} = \begin{bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \gamma_{xy} \end{bmatrix} = \boldsymbol{\nabla}_S \mathbf{u} = \boldsymbol{\nabla}_S \begin{bmatrix} u_x \\ u_y \end{bmatrix}, \tag{3}$$

  where $\boldsymbol{\nabla}_S$ is the symmetric gradient matrix operator

$$\boldsymbol{\nabla}_S = \begin{bmatrix} \partial/\partial x & 0 \\ 0 & \partial/\partial y \\ \partial/\partial y & \partial/\partial x \end{bmatrix}. \tag{4}$$

The kinematics relationships in 3D can be written as

$$\boldsymbol{\varepsilon} = \boldsymbol{\nabla}_S \mathbf{u},$$

$$\boldsymbol{\varepsilon} = \begin{bmatrix} \varepsilon_{xx} & \varepsilon_{yy} & \varepsilon_{zz} & \varepsilon_{xy} & \varepsilon_{xz} & \varepsilon_{yz} \end{bmatrix}^{\mathrm{T}}, \qquad \mathbf{u} = \begin{bmatrix} u_x & u_y & u_z \end{bmatrix}^{\mathrm{T}},$$

where

$$\boldsymbol{\nabla}_S^{\mathrm{T}} = \begin{bmatrix} \dfrac{\partial}{\partial x} & 0 & 0 & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} & 0 \\ 0 & \dfrac{\partial}{\partial y} & 0 & \dfrac{\partial}{\partial x} & 0 & \dfrac{\partial}{\partial z} \\ 0 & 0 & \dfrac{\partial}{\partial z} & 0 & \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} \end{bmatrix}.$$

Stress and Traction

- Stresses can be arranged in a matrix form similarly to strains:

$$\boldsymbol{\sigma}^{\mathrm{T}} = \begin{bmatrix} \sigma_{xx} & \sigma_{yy} & \sigma_{xy} \end{bmatrix}. \tag{7}$$

- Occasionally, it is convenient to arrange the stress components in a 2×2 symmetric matrix $\boldsymbol{\tau}$:

$$\boldsymbol{\tau} = \begin{bmatrix} \sigma_{xx} & \sigma_{xy} \\ \sigma_{xy} & \sigma_{yy} \end{bmatrix}. \tag{8}$$

- Consider the triangular body in Figure 9.4. The unit normal vector is $\vec{n} = n_x\vec{i} + n_y\vec{j}$. The traction components are

$$\begin{aligned} t_x &= \sigma_{xx} n_x + \sigma_{xy} n_y = \vec{\sigma}_x \cdot \vec{n}, \\ t_y &= \sigma_{xy} n_x + \sigma_{yy} n_y = \vec{\sigma}_y \cdot \vec{n}, \end{aligned} \tag{9}$$

which in matrix form reads

$$\mathbf{t} = \boldsymbol{\tau}\mathbf{n}. \tag{10}$$

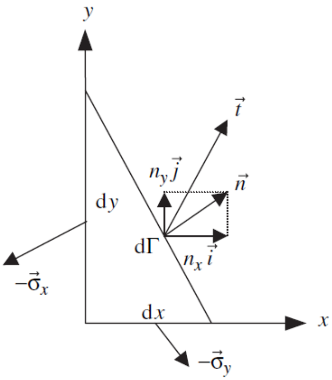

Figure 9.4 Relationship between stress and traction.

Figure 9.4 Relationship between stress and traction.

- Equilibrium equations:
$$\frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \sigma_{xy}}{\partial y} + b_x = 0, \qquad \frac{\partial \sigma_{yx}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} + b_y = 0. \tag{11}$$
- Or in the vector form:
$$\vec{\nabla} \cdot \vec{\sigma}_x + b_x = 0, \qquad \vec{\nabla} \cdot \vec{\sigma}_y + b_y = 0. \tag{12}$$
- The matrix form of the equilibrium equations can be written as
$$\boldsymbol{\nabla}_S^{\mathrm{T}} \boldsymbol{\sigma} + \mathbf{b} = \mathbf{0}. \tag{13}$$
- The selfadjointness (symmetry) of these partial differential equations is the underlying reason for the symmetry of the discrete equations.

- In two dimensions, the most general linear relation between the stress and strain matrices is

$$\boldsymbol{\sigma} = \mathbf{D}\boldsymbol{\varepsilon}, \tag{14}$$

  where $\mathbf{D}$ is a 3×3 matrix. This expression is called the **generalised Hooke's law**. It is always a symmetric, positive-definite matrix.

- Many materials, such as most steels, aluminiums, soil and concrete, are modelled as isotropic. For an isotropic material, the $\mathbf{D}$ matrix is given by:

Plane stress

$$\mathbf{D} = \frac{E}{1-\nu^2}\begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & (1-\nu)/2 \end{bmatrix}.$$

Plane strain

$$\mathbf{D} = \frac{E}{(1+\nu)(1-2\nu)}\begin{bmatrix} 1-\nu & \nu & 0 \\ \nu & 1-\nu & 0 \\ 0 & 0 & (1-2\nu)/2 \end{bmatrix}.$$

- For an isotropic material, the Hookean matrix $\mathbf{D}$ has two independent material constants: Young's modulus E and Poisson's ratio ν. For plane strain, as $\nu \to 0.5$, the Hookean matrix becomes infinite; a Poisson's ratio of 0.5 corresponds to an incompressible material.
- The Hookean matrix can also be written in terms of the bulk modulus $K = E/3(1-\nu)$ and the shear modulus $G = E/2(1+\nu)$.

Strain–displacement relation: $\boldsymbol{\varepsilon} = \boldsymbol{\nabla}_S \mathbf{u}$, $\boldsymbol{\varepsilon} = \begin{bmatrix} \varepsilon_{xx} & \varepsilon_{yy} & \varepsilon_{zz} & \varepsilon_{xy} & \varepsilon_{xz} & \varepsilon_{yz} \end{bmatrix}^{\mathrm{T}}$, $\mathbf{u} = \begin{bmatrix} u_x & u_y & u_z \end{bmatrix}^{\mathrm{T}}$.

Isotropic Hooke's law: $\boldsymbol{\sigma} = \mathbf{D}\boldsymbol{\varepsilon}$,

$$\mathbf{D} = \frac{E}{(1+\nu)(1-2\nu)} \begin{bmatrix} 1-\nu & \nu & \nu & 0 & 0 & 0 \\ \nu & 1-\nu & \nu & 0 & 0 & 0 \\ \nu & \nu & 1-\nu & 0 & 0 & 0 \\ 0 & 0 & 0 & \dfrac{1-2\nu}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & \dfrac{1-2\nu}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \dfrac{1-2\nu}{2} \end{bmatrix}.$$

- Strong form for linear elasticity:

$$
\begin{aligned}
&\text{(a)} \quad \vec{\nabla} \cdot \vec{\sigma}_x + b_x = 0 \quad \text{and} \quad \vec{\nabla} \cdot \vec{\sigma}_y + b_y = 0 \quad \text{on} \quad \Omega,\\
&\text{(b)} \quad \boldsymbol{\sigma} = \mathbf{D}\boldsymbol{\nabla}_S \mathbf{u},\\
&\text{(c)} \quad \vec{\sigma}_x \cdot \vec{n} = \bar{t}_x \quad \text{and} \quad \vec{\sigma}_y \cdot \vec{n} = \bar{t}_y \quad \text{on} \quad \Gamma_t,\\
&\text{(d)} \quad \vec{u} = \bar{\vec{u}} \quad \text{on} \quad \Gamma_u.
\end{aligned}
\tag{15}
$$

- To obtain the weak form, we premultiply the equilibrium equations (15a) and the natural boundary conditions (15c) by the corresponding weight functions and integrate over the corresponding domains, which gives

$$
\begin{aligned}
&\text{(a)} \quad \int_\Omega w_x \, \vec{\nabla} \cdot \vec{\sigma}_x \, \mathrm{d}\Omega + \int_\Omega w_x b_x \, \mathrm{d}\Omega = 0 \qquad \forall w_x \in U_0,\\
&\text{(b)} \quad \int_\Omega w_y \, \vec{\nabla} \cdot \vec{\sigma}_y \, \mathrm{d}\Omega + \int_\Omega w_y b_y \, \mathrm{d}\Omega = 0 \qquad \forall w_y \in U_0,\\
&\text{(c)} \quad \int_{\Gamma_t} w_x (\bar{t}_x - \vec{\sigma}_x \cdot \vec{n}) \, \mathrm{d}\Gamma = 0 \qquad \forall w_x \in U_0,\\
&\text{(d)} \quad \int_{\Gamma_t} w_y (\bar{t}_y - \vec{\sigma}_y \cdot \vec{n}) \, \mathrm{d}\Gamma = 0 \qquad \forall w_y \in U_0,
\end{aligned}
\tag{16}
$$

where $\mathbf{w} = \begin{bmatrix} w_x \\ w_y \end{bmatrix}, \quad \vec{w} = w_x \vec{i} + w_y \vec{j}.$

- Green's theorem is applied to the first terms in equations (16a) and (16b), which yields

$$
\begin{aligned}
\int_\Omega w_x\, \vec{\nabla} \cdot \vec{\sigma}_x \,\mathrm{d}\Omega &= \oint_\Gamma w_x \vec{\sigma}_x \cdot \vec{n}\,\mathrm{d}\Gamma - \int_\Omega \vec{\nabla} w_x \cdot \vec{\sigma}_x \,\mathrm{d}\Omega, \\
\int_\Omega w_y\, \vec{\nabla} \cdot \vec{\sigma}_y \,\mathrm{d}\Omega &= \oint_\Gamma w_y \vec{\sigma}_y \cdot \vec{n}\,\mathrm{d}\Gamma - \int_\Omega \vec{\nabla} w_y \cdot \vec{\sigma}_y \,\mathrm{d}\Omega.
\end{aligned}
\tag{17}
$$

- Adding the two equations in (17) and recalling that w_x and w_y vanish on Γ_u yields

$$
\int_\Omega (\vec{\nabla} w_x \cdot \vec{\sigma}_x + \vec{\nabla} w_y \cdot \vec{\sigma}_y)\mathrm{d}\Omega = \oint_{\Gamma_t} (w_x \vec{\sigma}_x \cdot \vec{n} + w_y \vec{\sigma}_y \cdot \vec{n})\,\mathrm{d}\Gamma + \int_\Omega (w_x b_x + w_y b_y)\,\mathrm{d}\Omega. \tag{18}
$$

- Substituting (16c) and (16d) into (18) and writing the RHS in vector form gives

$$
\int_\Omega (\vec{\nabla} w_x \cdot \vec{\sigma}_x + \vec{\nabla} w_y \cdot \vec{\sigma}_y)\mathrm{d}\Omega = \oint_{\Gamma_t} \vec{w} \cdot \vec{t}\,\mathrm{d}\Gamma + \int_\Omega \vec{w} \cdot \vec{b}\,\mathrm{d}\Omega. \tag{19}
$$

- Expanding the integrand on the LHS of (19) yields

$$
\vec{\nabla} w_x \cdot \vec{\sigma}_x + \vec{\nabla} w_y \cdot \vec{\sigma}_y = \begin{bmatrix} \dfrac{\partial w_x}{\partial x} & \dfrac{\partial w_y}{\partial y} & \left(\dfrac{\partial w_x}{\partial y} + \dfrac{\partial w_y}{\partial x} \right) \end{bmatrix} \begin{bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \sigma_{xy} \end{bmatrix} = (\boldsymbol{\nabla}_S \mathbf{w})^{\mathrm{T}} \boldsymbol{\sigma}. \tag{20}
$$

- Inserting (20) into (19) and writing the RHS of (19) in matrix form gives

$$\int_{\Omega} (\boldsymbol{\nabla}_S \mathbf{w})^{\mathrm{T}} \boldsymbol{\sigma} \, \mathrm{d}\Omega = \int_{\Gamma_t} \mathbf{w}^{\mathrm{T}} \bar{\mathbf{t}} \, \mathrm{d}\Gamma + \int_{\Omega} \mathbf{w}^{\mathrm{T}} \mathbf{b} \, \mathrm{d}\Omega \qquad \forall \mathbf{w} \in U_0. \tag{21}$$

- After the substitution of (15b) for $\boldsymbol{\sigma}$, the weak form in two dimensions is:

Weak form

Find $\mathbf{u} \in U$ such that

$$\int_{\Omega} (\boldsymbol{\nabla}_S \mathbf{w})^{\mathrm{T}} \mathbf{D} \boldsymbol{\nabla}_S \mathbf{u} \, \mathrm{d}\Omega = \int_{\Gamma_t} \mathbf{w}^{\mathrm{T}} \bar{\mathbf{t}} \, \mathrm{d}\Gamma + \int_{\Omega} \mathbf{w}^{\mathrm{T}} \mathbf{b} \, \mathrm{d}\Omega \qquad \forall \mathbf{w} \in U_0, \tag{22}$$

where $U = \{\mathbf{u} \mid \mathbf{u} \in H^1, \ \mathbf{u} = \bar{\mathbf{u}} \text{ on } \Gamma_u\}, \quad U_0 = \{\mathbf{w} \mid \mathbf{w} \in H^1, \ \mathbf{w} = \mathbf{0} \text{ on } \Gamma_u\}.$

Finite Element Discretisation

- Consider a problem domain Ω with boundary Γ discretised with two-dimensional elements (triangles or quadrilaterals). The total number of elements is n_{el}.
- There are **two degrees of freedom per node** corresponding to the two components of the global displacements, so the nodal displacement matrix is:

$$\mathbf{d} = \begin{bmatrix} u_{x1} & u_{y1} & u_{x2} & u_{y2} & \cdots & u_{xn_{\mathrm{np}}} & u_{yn_{\mathrm{np}}} \end{bmatrix}^{\mathrm{T}},$$

  where n_{np} is the number of nodes in the mesh.
- The finite element approximation of the trial solution and weight function on each element is:

$$\begin{aligned} \mathbf{u}(x,y) &\approx \mathbf{u}^e(x,y) = \mathbf{N}^e(x,y)\mathbf{d}^e, \quad (x,y) \in \Omega^e, \\ \mathbf{w}^{\mathrm{T}}(x,y) &\approx \mathbf{w}^{e\mathrm{T}}(x,y) = \mathbf{w}^{e\mathrm{T}}\mathbf{N}^e(x,y)^{\mathrm{T}}, \quad (x,y) \in \Omega^e. \end{aligned} \tag{23}$$

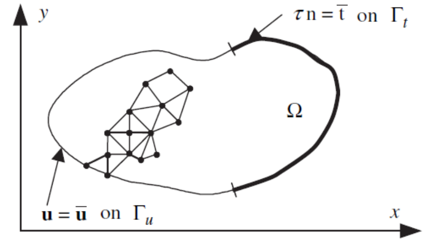

Figure 9.6 Finite element mesh in two dimensions.

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- The element shape function matrix in (23) is given as

$$\mathbf{N}^e = \begin{bmatrix} N_1^e & 0 & N_2^e & 0 & \cdots & N_{n_{\mathrm{en}}}^e & 0 \\ 0 & N_1^e & 0 & N_2^e & \cdots & 0 & N_{n_{\mathrm{en}}}^e \end{bmatrix}, \tag{24}$$

and $\mathbf{d}^e = \begin{bmatrix} u_{x1}^e & u_{y1}^e & u_{x2}^e & u_{y2}^e & \cdots & u_{xn_{\mathrm{en}}}^e & u_{yn_{\mathrm{en}}}^e \end{bmatrix}^{\mathrm{T}}$ are the element nodal displacements, while $\mathbf{w}^e = \begin{bmatrix} w_{x1}^e & w_{y1}^e & w_{x2}^e & w_{y2}^e & \cdots & w_{xn_{\mathrm{en}}}^e & w_{yn_{\mathrm{en}}}^e \end{bmatrix}^{\mathrm{T}}$ are the element nodal weight functions.
- The integral over Ω in the weak form (22) is computed as a sum of integrals over element domains:

$$\sum_{e=1}^{n_{\mathrm{el}}} \left\{ \int_{\Omega^e} \boldsymbol{\nabla}_S \mathbf{w}^{e\mathrm{T}} \mathbf{D}^e \boldsymbol{\nabla}_S \mathbf{u}^e \,\mathrm{d}\Omega - \int_{\Gamma_t^e} \mathbf{w}^{e\mathrm{T}} \bar{\mathbf{t}} \,\mathrm{d}\Gamma - \int_{\Omega^e} \mathbf{w}^{e\mathrm{T}} \mathbf{b} \,\mathrm{d}\Omega \right\} = 0. \tag{24}$$

- Applying the symmetric gradient operator to $\mathbf{N}^e$ gives the strain–displacement matrix:

$$\boldsymbol{\varepsilon} \approx \boldsymbol{\varepsilon}^e = \boldsymbol{\nabla}_S \mathbf{u}^e = \boldsymbol{\nabla}_S \mathbf{N}^e \mathbf{d}^e = \mathbf{B}^e \mathbf{d}^e, \tag{25}$$

where the strain–displacement matrix $\mathbf{B}^e$ is defined as

$$\mathbf{B}^e \equiv \boldsymbol{\nabla}_S \mathbf{N}^e = \begin{bmatrix} \dfrac{\partial N_1^e}{\partial x} & 0 & \dfrac{\partial N_2^e}{\partial x} & 0 & \cdots & \dfrac{\partial N_{n_{\mathrm{en}}}^e}{\partial x} & 0 \\ 0 & \dfrac{\partial N_1^e}{\partial y} & 0 & \dfrac{\partial N_2^e}{\partial y} & \cdots & 0 & \dfrac{\partial N_{n_{\mathrm{en}}}^e}{\partial y} \\ \dfrac{\partial N_1^e}{\partial y} & \dfrac{\partial N_1^e}{\partial x} & \dfrac{\partial N_2^e}{\partial y} & \dfrac{\partial N_2^e}{\partial x} & \cdots & \dfrac{\partial N_{n_{\mathrm{en}}}^e}{\partial y} & \dfrac{\partial N_{n_{\mathrm{en}}}^e}{\partial x} \end{bmatrix}.$$

The derivatives of the weight functions are:

$$(\boldsymbol{\nabla}_S \mathbf{w}^e)^{\mathrm{T}} = (\mathbf{B}^e \mathbf{w}^e)^{\mathrm{T}} = \mathbf{w}^{e\mathrm{T}} \mathbf{B}^{e\mathrm{T}}. \tag{26}$$

- The element matrices are given as follows.

Element stiffness matrix

$$\mathbf{K}^e = \int_{\Omega^e} (\mathbf{B}^e)^{\mathrm{T}} \mathbf{D}^e \mathbf{B}^e \, \mathrm{d}\Omega. \tag{28}$$

Element external force matrix

$$\mathbf{f}^e = \underbrace{\int_{\Omega^e} (\mathbf{N}^e)^{\mathrm{T}} \mathbf{b} \, \mathrm{d}\Omega}_{\mathbf{f}^e_\Omega} + \underbrace{\int_{\Gamma^e_t} (\mathbf{N}^e)^{\mathrm{T}} \bar{\mathbf{t}} \, \mathrm{d}\Gamma}_{\mathbf{f}^e_\Gamma}, \tag{29}$$

where $\mathbf{f}^e_\Omega$ and $\mathbf{f}^e_\Gamma$ are the body and boundary force matrices.

- The weak form can then be written as

$$\mathbf{w}^{\mathrm{T}} \left[\left(\sum_{e=1}^{n_{\mathrm{el}}} (\mathbf{L}^e)^{\mathrm{T}} \mathbf{K}^e \mathbf{L}^e \right) \mathbf{d} - \sum_{e=1}^{n_{\mathrm{el}}} (\mathbf{L}^e)^{\mathrm{T}} \mathbf{f}^e \right] = 0 \qquad \forall \mathbf{w}_F, \tag{30}$$

where $\mathbf{K} = \sum_{e=1}^{n_{\mathrm{el}}} (\mathbf{L}^e)^{\mathrm{T}} \mathbf{K}^e \mathbf{L}^e$ and $\mathbf{f} = \sum_{e=1}^{n_{\mathrm{el}}} (\mathbf{L}^e)^{\mathrm{T}} \mathbf{f}^e$.

- The system (30) can be reduced to
$$\mathbf{w}^{\mathrm{T}}(\mathbf{K}\mathbf{d}-\mathbf{f})=0 \qquad \forall \mathbf{w}_F. \tag{31}$$
- Equation (31) can be written as
$$\mathbf{w}^{\mathrm{T}}\mathbf{r}=0 \qquad \forall \mathbf{w}_F. \tag{32}$$
- Partitioning equation (32) into E- and F-nodes gives
$$\mathbf{w}_F^{\mathrm{T}}\mathbf{r}_F+\mathbf{w}_E^{\mathrm{T}}\mathbf{r}_E=0 \qquad \forall \mathbf{w}_F.$$

  As $\mathbf{w}_E=\mathbf{0}$ and $\mathbf{w}_F$ is arbitrary, it follows that $\mathbf{r}_F=\mathbf{0}$. Consequently, the above equation can be conveniently rewritten as
$$\begin{bmatrix}\mathbf{K}_E & \mathbf{K}_{EF}\\ \mathbf{K}_{EF}^{\mathrm{T}} & \mathbf{K}_F\end{bmatrix}\begin{bmatrix}\bar{\mathbf{d}}_E\\ \mathbf{d}_F\end{bmatrix}=\begin{bmatrix}\mathbf{f}_E+\mathbf{r}_E\\ \mathbf{f}_F\end{bmatrix}, \tag{33}$$
  where $\mathbf{K}_E$, $\mathbf{K}_F$ and $\mathbf{K}_{EF}$ are partitioned to be congruent with the partition of $\mathbf{d}$ and $\mathbf{f}$.

Next: Approximations of Trial Solutions for 2D and 3D Problems – Part 2

Thanks for your attention!